

Default Values in GUI Widgets

In the next section, I'll present a collection of classes that you can use to easily create GUI elements such as buttons and text fields within pygame. These classes will each be initialized using a few positional parameters but will also have assorted optional keyword parameters, all with reasonable defaults to allow programmers to create GUI widgets by specifying only a few positional arguments. More precise control can be obtained by specifying values to overwrite the default values of keyword parameters.

For an in-depth example, we'll look at a widget to display text in the application's window. Text can be shown in a variety of fonts, font sizes, colors, background colors, and so on. We'll build a `DisplayText` class that will have default values for all of these attributes but will give client code the option of specifying different values.

The pygwidgets Package

The rest of this chapter will focus on the `pygwidgets` (pronounced “pig wijits”) package, which was written with two goals in mind:

1. To demonstrate many different object-oriented programming techniques
2. To allow programmers to easily create and use GUI widgets in pygame programs

The `pygwidgets` package contains the following classes:

TextButton

Button built with standard art, using a text string

CustomButton

Button with custom artwork